

MARIINSK, Russian Federation

WMO: 295510

Lat: 56.183N

Lon: 87.683E

Elev: 144

StdP: 99.61

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -37.8	(c) -33.9	(d) -40.6	(e) 0.1	(f) -37.7	(g) -36.5	(h) 0.1	(i) -33.7	(j) 11.4	(k) -9.2	(l) 9.4	(m) -9.5	(n) 0.4	(o) 60	(p) 0.598

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.5	(c) 29.2	(d) 19.4	(e) 27.4	(f) 18.6	(g) 25.7	(h) 17.9	(i) 21.2	(j) 26.4	(k) 20.1	(l) 25.2	(m) 19.0	(n) 24.1	(o) 2.2	(p) 100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	14.5	23.8	18.2	13.3	22.7	17.1	12.4	21.5	62.5	26.4	58.4	25.2	54.8	24.3	26.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.5	(b) 8.2	(c) 7.0	(e) -40.5	(f) 32.3	(g) 4.0	(h) 1.4	(i) -43.4	(j) 33.3	(k) -45.7	(l) 34.1	(m) -47.9	(n) 34.9	(o) -50.8	(p) 35.9		
(a) 9.5	(b) 8.2	(c) 7.0	(e) -40.5	(f) 23.3	(g) 3.7	(h) 1.8	(i) -43.2	(j) 24.6	(k) -45.4	(l) 25.6	(m) -47.4	(n) 26.6	(o) -50.1	(p) 27.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.6	-17.3	-13.6	-5.7	3.1	10.3	16.7	18.7	15.9	9.4	2.4	-7.5	-14.1
	DBStd	13.91	9.25	7.82	6.43	5.45	5.39	4.18	3.15	3.70	4.51	5.41	8.37	8.93
	HDD10.0	3858	846	662	485	213	66	4	0	3	66	239	526	748
	HDD18.3	6203	1104	896	743	457	255	77	33	94	269	492	776	1007
	CDD10.0	793	0	0	0	7	73	206	271	184	47	5	0	0
	CDD18.3	96	0	0	0	0	4	28	46	17	1	0	0	0
	CDH23.3	1004	0	0	0	0	84	321	411	166	23	0	0	0
	CDH26.7	228	0	0	0	0	21	81	90	32	4	0	0	0
Wind	WSAvg	2.6	2.6	2.7	3.0	3.1	3.0	2.2	1.7	1.8	2.3	2.8	3.0	3.0
Precipitation	PrecAvg	441	18	12	14	28	43	62	60	64	44	38	34	25
	PrecMax	665	46	33	33	64	81	136	204	141	104	75	78	65
	PrecMin	292	3	2	2	3	12	14	10	9	12	12	9	4
	PrecStd	86	10	7	7	16	19	34	42	29	25	18	15	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.9	2.6	9.9	20.7	29.6	31.5	31.1	30.1	26.6	18.3	9.1	2.7
		MCWB	0.2	0.3	5.5	11.1	16.7	18.8	21.1	20.5	16.4	10.5	6.1	0.8
	2%	DB	-0.8	0.5	6.5	17.6	25.7	29.0	29.1	27.1	22.2	15.1	6.5	0.6
		MCWB	-2.2	-1.3	2.7	9.7	14.3	18.7	20.5	19.3	14.7	9.5	4.3	-0.8
	5%	DB	-2.6	-1.0	4.6	14.8	23.0	27.0	27.3	25.1	19.5	12.5	4.1	-1.0
		MCWB	-3.8	-2.4	1.4	8.3	13.1	17.6	19.4	18.1	13.4	8.2	2.5	-2.1
	10%	DB	-4.7	-2.9	3.0	12.0	20.1	25.0	25.6	23.1	16.9	9.9	2.2	-2.9
		MCWB	-5.8	-4.0	0.5	6.7	12.2	17.1	18.9	17.2	12.1	7.1	0.7	-3.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.3	0.8	5.4	12.3	17.8	21.6	23.4	22.5	17.1	11.9	7.0	1.2
		MCDB	1.7	2.3	9.4	17.5	26.8	27.1	27.9	27.4	23.6	15.8	8.9	2.4
	2%	WB	-2.0	-1.0	3.4	10.5	16.1	20.1	21.6	20.2	15.3	10.1	4.4	-0.6
		MCDB	-0.9	0.3	5.7	16.8	23.5	25.7	26.6	25.1	21.0	14.3	5.9	0.4
	5%	WB	-3.8	-2.5	1.8	8.7	14.3	19.0	20.6	18.9	14.0	8.6	2.6	-2.1
		MCDB	-2.6	-1.2	4.1	14.0	20.5	25.1	25.7	23.6	18.3	12.1	4.0	-1.0
	10%	WB	-5.7	-4.0	0.5	6.9	12.8	18.0	19.6	17.8	12.7	7.1	0.7	-3.9
		MCDB	-4.6	-2.9	2.8	11.9	19.1	23.6	24.4	22.2	16.2	9.9	2.0	-2.8
Mean Daily Temperature Range	5% DB	MDBR	8.0	9.0	9.5	9.7	11.2	10.9	10.5	10.3	9.7	7.2	6.5	7.5
		MCDBR	7.3	7.5	10.2	14.4	16.1	14.9	13.5	13.8	14.8	11.9	7.0	7.8
		MCWBR	6.5	6.5	7.4	8.2	7.3	6.4	5.9	6.7	7.8	7.2	5.4	7.2
	5% WB	MCDBR	7.1	7.2	9.5	13.8	14.1	12.5	11.5	12.0	12.5	10.8	6.5	8.1
		MCWBR	6.5	6.5	7.4	8.6	7.3	6.4	6.1	6.4	7.0	7.1	5.3	7.5
Clear-Sky Solar Irradiance	taub	0.291	0.325	0.374	0.404	0.412	0.408	0.436	0.411	0.373	0.355	0.323	0.293	
	taud	1.994	1.968	1.904	2.055	2.190	2.292	2.213	2.262	2.314	2.215	2.032	1.945	
	Ebn at Noon	557	693	753	804	826	835	799	794	772	669	507	434	
	Edh at Noon	67	103	142	143	133	122	130	116	95	81	65	55	
All-Sky Solar Radiation	RadAvg	0.79	1.78	3.14	4.36	4.91	5.83	5.29	4.24	2.77	1.39	0.68	0.47	
	RadStd	0.09	0.14	0.26	0.32	0.55	0.55	0.35	0.33	0.36	0.13	0.11	0.08	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(47) Regional (12 neighbors)	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A

Nomenclature: See separate page